Course no: ME-498

# PHYSICS-INFORMED NEURAL NETWORKS FOR REMAINING-USEFUL-LIFE PREDICTION OF ROLLING-ELEMENT BEARINGS

A thesis submitted in partial fulfillment of
requirements for the degree of

Bachelor of Science

in

Mechanical Engineering

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**CHITTAGONG UNIVERSITY OF ENGINEERING AND TECHNOLOGY, CHATTOGRAM**

(PROJECT AND THESIS)

**COURSE NO: ME-498**

**Title of Project:** Physics-Informed Neural Networks for Remaining-Useful-Life Prediction of Rolling-Element Bearings

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# Abstract

The condition of rolling element bearings is often a critical factor in determining the safety in operation of a rotating machine, despite being a small component. A bearing fault can start out as a localized surface fault, escalate to create vibration, heat, misalignment, secondary damage, and unplanned shutdown. Prediction of the remaining useful life is thus a key issue in the field of prognostics and health management.. This thesis studies bearing remaining useful life prediction using the NASA IMS run-to-failure bearing data and compares a physics-informed neural network against three data-driven baselines. The original proposal for this work expected a physics-informed model to outperform a purely data-driven model. We got more measured results from the local experiments which were completed. The proposed DeepXDE physics-informed model performed best in one of the three cross-bearing experiments, but the LSTM baseline gave the lowest average RMSE and highest average $R^2$ across the full experiment set. This does not make the physics-informed approach unimportant. It shows that weak physical regularization, when applied to noisy run-to-failure vibration features, must be designed and weighted carefully. A simple monotonicity and fault-frequency prior can help in some transfer settings, but it may also restrict the network when the degradation path of the test bearing differs from the training bearings. The final pipeline reads the IMS data locally from extracted folders, removes the dependency on Google Drive, extracts time-domain and envelope spectral features, constructs normalized RUL targets, builds a PCA-based health indicator, and evaluates four models: a feed-forward neural baseline, the proposed DeepXDE physics-informed model, an LSTM sequence model, and a 1D CNN sequence model. The experiments use three train-validation-test splits designed to test cross-bearing and cross-dataset transfer. The best average RMSE was achieved by the LSTM baseline at 0.162, followed by the data-only feed-forward baseline at 0.185, the proposed physics-informed model at 0.223, and the CNN baseline at 0.235. The work contributes a reproducible local project structure, a documented feature and evaluation pipeline, and an honest analysis of where the investigated physics-informed formulation helped and where it did not.

**Keywords:** remaining useful life, rolling-element bearing, prognostics and health management, physics-informed neural network, IMS bearing dataset, vibration analysis, LSTM, CNN.

# Acknowledgement

First, I offer my gratitude to Allah for giving me the strength, patience, and opportunity to complete this thesis work. I am deeply grateful to my undergraduate thesis supervisor, Dr. Md. Abu Mowazzem Hossain, Professor, Department of Mechanical Engineering, Chittagong University of Engineering & Technology, for his supervision, advice, and encouragement throughout this project.

I also thank my family for their constant support during the long process of learning the theory, writing code, running experiments, correcting mistakes, and preparing this thesis. Their patience made the work easier to continue when the results did not come out as simply as expected.

Finally, I acknowledge the researchers who made the IMS bearing dataset publicly available. Open run-to-failure datasets make it possible for students and researchers to test prognostic methods in a reproducible way.

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# Nomenclature

| Symbol or term | Meaning |
|---|---|
| ANN | Artificial neural network |
| BPFO | Ball pass frequency outer race |
| BPFI | Ball pass frequency inner race |
| BSF | Ball spin frequency |
| CNN | Convolutional neural network |
| DeepXDE | Python library used for physics-informed neural networks |
| FFT | Fast Fourier transform |
| FNN | Feed-forward neural network |
| FTF | Fundamental train frequency |
| HI | Health indicator |
| IMS | Intelligent Maintenance Systems |
| LSTM | Long short-term memory network |
| MAE | Mean absolute error |
| PCA | Principal component analysis |
| PHM | Prognostics and health management |
| PINN | Physics-informed neural network |
| REB | Rolling-element bearing |
| RMS | Root mean square |
| RMSE | Root mean squared error |
| RUL | Remaining useful life |
| $R^2$ | Coefficient of determination |

# Chapter 1

# Introduction

## 1.1 BACKGROUND AND MOTIVATION

Rolling-element bearings support shafts, carry loads, and reduce friction in motors, pumps, turbines, gearboxes, fans, compressors, and many other machines. Because they sit directly in the load path, a bearing defect rarely remains an isolated issue for long. A small defect on a raceway or rolling element can create repeated impacts, excite vibration, produce heat, damage nearby parts, and eventually force the machine out of service. The consequences of a bearing failure are sometimes far greater than the cost of the bearing in industrial maintenance.

While traditional maintenance approaches could help, they don't completely address the issue. Reactive maintenance is waiting for failure and takes the risk of unplanned shutdown. Preventive maintenance involves exchanges of parts at a predetermined interval, typically a more conservative interval than necessary. Can take out healthy components too soon or miss faults that occur quicker than they should. The improvements of both methods are provided in condition-based maintenance, in that the state of the machine is assessed and a maintenance decision is made based on the evidence. Prognostics and health management is a further step that attempts to predict the remaining useful life of a component before failure, in addition to predicting failure, [2, 3].

The reason why the remaining useful life prediction for bearings is difficult is that the deterioration process is not linear. Vibration may stay low for a considerable length of the test, but then increase dramatically towards the end of life. The measured signals are affected by the type of fault, the speed of the fault, the load, the position of the sensor, noise, resonance, lubrication, and the geometry of the bearings [4, 5]. Training on one of the bearings doesn't mean that the model works well on another bearing, even if they are both from the same test rig. This is the reason that the IMS bearing data set is helpful for benchmarking run-to-failure experiments. They enable a study of the same raw measurements and the comparison of methods under identical conditions [1].

The deep learning techniques have gained popularity for their ability to capture nonlinear patterns within vibration signals, which is crucial for bearing diagnosis and RUL prediction. CNNs can learn local patterns from windows of data, LSTMs can model temporal dependence, and feed-forward networks can fit engineered feature vectors [6–8]. These methods can work well when the training and test data are similar. Their weakness is that they usually learn only from examples. When labeled run-to-failure data are limited or the test bearing degrades differently from the training bearings, a data-only model may produce predictions that are numerically plausible but physically questionable.

Physics-informed neural networks offer a way to include prior engineering knowledge in the learning process. In their most established form, PINNs train a neural network while penalizing violations of governing equations or physical constraints [9]. Bearing RUL prediction does not usually provide a complete closed-form degradation equation, but it does provide useful partial knowledge. Damage should generally increase with operating time, RUL should generally decrease, and bearing geometry gives characteristic fault frequencies associated with cage, inner-race, outer-race, and rolling-element defects. This thesis studies whether such weak physical information can improve RUL prediction on the IMS data.

## 1.2 PROBLEM STATEMENT

The problem studied in this thesis is the prediction of normalized remaining useful life for selected bearings in the IMS run-to-failure dataset. The main question is whether a physics-informed neural network using time and fault-frequency information can generalize better across bearing runs than comparable data-driven neural baselines.

This question is practical as well as methodological. In a real machine, the future failure path is unknown. A maintenance engineer may have historical data from one bearing, but the next bearing may not degrade in the same way. A useful RUL model must therefore do more than memorize one run. It must learn a relationship between measured condition and remaining life that transfers, at least partly, across bearings.

## 1.3 OBJECTIVES OF THE STUDY

The objectives of this thesis are:

1. To extract degradation-relevant vibration features from selected IMS bearing runs and construct normalized RUL targets with a PCA-based health indicator for analysis.

2. To design and evaluate a DeepXDE physics-informed neural model that uses weak monotonicity and fault-frequency energy priors for bearing RUL prediction.

3. To compare the proposed physics-informed model with data-only, LSTM, and CNN

neural baselines under complete held-out bearing splits and interpret where each model succeeds or fails.

## 1.4 SCOPE

This study is limited to the IMS bearing data placed locally under `data/raw/1st_test`, `data/raw/2nd_test`, and `data/raw/3rd_test`. Four bearing runs are used in the final experiment design: `ds2_b1`, `ds1_b3`, `ds1_b4`, and `ds3_b3`. The work uses vibration signals only. It does not add temperature, oil debris, acoustic emission, or motor-current data.

The models predict normalized RUL from extracted features. The thesis does not claim field deployment readiness. The data come from a controlled test rig, and the operational conditions are much simpler than many industrial machines. There is also an intentional weak physical information from PINN. It relies on monotonicity and energy priors for fault frequency, instead of a full energy balance contact mechanics or crack growth model.

## 1.5 IMPORTANCE OF THE STUDY

This work is the integration of three parts of the bearing prognostics problem in one local workflow: Signal processing, machine learning and physical interpretation. The feature extraction step maintains the links with vibration analysis of RMS, kurtosis, crest factor, and envelope spectral energies. The neural models investigate if those features can contribute to Predicting RULs over bearing runs. The physics-informed model is used to test if simple Priorities can be regularized in the learning process through physical priors. The final result isn't just a success story for the proposed model. That is important. A thesis result should state what happens during the experiment, not what is expected to happen in the proposal. The final run indicates that LSTM sequence modeling was able to capture the degradation .The proposed weakly physics-informed feed-forward model is less consistent in their trend. Meanwhile, the PINN was competitive in Experiment 2 and ranked first in Experiment 2 (RMSE).This mixed behavior is more clearly specified for future work than an overconfident positive claim would.

## 1.6 LIMITATIONS

The main limitations are:

1. The IMS dataset has small number of total run-to-failure records which creates a experiment design constraint.

2. The physics selected prior is not a full bearing degradation law. It is a regularizer based on monotonic RUL behavior and fault-frequency energy.

3. The model uses hand-engineered features, not raw vibration waveform.

4. The RUL labels are derived from elapsed time to the end of each run, which assumes the last available file represents failure or near-failure.

5. The experiments are run on one local PC, so training time and numerical results may vary on other machines.

## 1.7 THESIS OUTLINE

Chapter 2 reviews bearing failure, vibration-based condition monitoring, RUL prediction, deep learning baselines, and physics-informed learning. Chapter 3 describes the IMS dataset, feature extraction, preprocessing, health indicator construction, model architectures, and evaluation metrics. Chapter 4 describes the local implementation, experiment splits, software environment, and reproducibility setup. Chapter 5 presents the results and discusses model behavior across the three experiments. Chapter 6 summarizes the findings, contributions, limitations, and future work.

# Chapter 2

# Literature Review

## 2.1 INTRODUCTION

This chapter reviews the research background for bearing RUL prediction with physics-informed neural networks. The discussion follows the structure of the thesis proposal, but it is updated to reflect the completed experiments. It begins with rolling-element bearing failure modes and vibration-based health monitoring, then reviews predictive maintenance methods, deep learning for bearing prognosis, general PINN theory, and recent physics-informed learning work in bearing condition monitoring. The chapter closes by identifying the research gap addressed in this thesis.

## 2.2 LITERATURE SURVEY

### 2.2.1 Rolling-Element Bearings: Fundamentals and Failure Modes

A rolling-element bearing contains an inner race, an outer race, rolling elements, and a cage. The rolling elements transfer load through small contact regions, so repeated stress cycles can initiate subsurface or surface-initiated fatigue. Defects may also arise from lubrication failure, contamination, overload, misalignment, electrical pitting, poor installation, or manufacturing variation. Once a defect begins, repeated rolling contact can grow it into a spall or crack that changes the vibration response of the machine [10–12].

The vibration signature depends on the component that is damaged. An outer-race defect produces impacts when rolling elements pass over a fixed damaged region. An inner-race defect rotates with the shaft and is affected by the load zone. Rolling-element and cage defects produce different modulation patterns. These mechanisms explain why bearing monitoring often uses characteristic frequencies derived from bearing geometry and shaft speed. The frequencies are not ideal fault labels but they define meaningful regions in the spectrum, and envelope spectrum [4, 13].

The thesis proposal highlighted that the vibration features are not arbitrary numbers, but a legitimate representation of the data and hence should be used as such in bearing prognostics. This is still relevant in the finished product. Even though these are subsequently used in a neural model, RMS, kurtosis, crest factor, and envelope energy all have a mechanical meaning. Physically interpretable features also enable a deeper understanding of why a model is successful or unsuccessful in a held-out run of bearings.

### 2.2.2 Traditional Predictive Maintenance Techniques for Bearings

Normally, predictive maintenance is part of condition-based maintenance and prognostics and health management. Jardine et al. [2] discuss the overall machinery diagnostics and prognostics, and Lei et al. [3] review the entire process of data acquisition to RUL prediction. The typical process for a bearing application covers sensor measurement, preprocessing, feature extraction, construction of health indicators, diagnosis of the faults and estimation of future life.

Vibration remains one of the most widely used signals for bearing health monitoring because bearing faults generate impacts and modulations in the mechanical response. Time-domain features such as RMS and kurtosis summarize energy and impulsiveness. Frequency-domain and envelope-domain features can reveal periodic impact patterns linked to bearing kinematics. Similarity-based and health-indicator methods then convert these features into a degradation measure or RUL estimate [14].

The IMS bearing dataset is one of the common public run-to-failure datasets used for this type of work. It is associated with the wavelet-filter bearing prognostics study of Qiu et al. [1]. The PRONOSTIA platform is another well-known accelerated bearing degradation benchmark [15]. Public datasets are valuable because they let different researchers test methods on the same raw measurements. At the same time, public datasets are small compared with industrial variability, so the train-test split must be designed carefully. Random snapshot splits can overstate performance by allowing information from the same run-to-failure trajectory into both training and testing.

### 2.2.3 Deep Learning in Predictive Maintenance

Deep learning has become popular in bearing prognostics because it can fit nonlinear relationships between sensor measurements and degradation labels. A feed-forward neural network can work directly with engineered feature vectors. A CNN can detect local patterns from a sequence or transformed signal representation. An LSTM can model temporal dependence and is therefore naturally suited to degradation trajectories. Babu et al. [6] showed early CNN-based RUL regression, and later studies extended deep learning for transfer learning, high-quality representation learning, and hybrid sequence architectures [7, 8, 16].

Recent studies also combine convolutional, recurrent, and autoencoder structures for bearing health monitoring and RUL prediction [17–19]. These models are attractive because inference is fast after training and because they can learn patterns that are difficult to specify manually. However, their performance depends strongly on the similarity between the training and test distributions. In bearing RUL prediction, this is a serious issue because two bearings under similar nominal conditions may fail at different times and with different feature trajectories.

For this reason, model evaluation should hold out complete bearing runs when the research question is transfer across bearings. A random split of individual snapshots may measure interpolation within one degradation curve rather than generalization to another bearing. The experiment design in this thesis uses complete held-out runs for validation and testing.

### 2.2.4 Physics-Informed Neural Networks

Physics-informed neural networks were introduced as a way to train neural networks under physical residual constraints [9]. In the usual PINN setting, a neural network approximates a field variable, and automatic differentiation is used to compute derivatives that appear in a governing equation. The training loss combines data mismatch and a physics residual. This has made PINNs attractive in fluid mechanics, heat transfer, wave propagation, and inverse problems where governing equations are available.

The body of PINN literature has been growing rapidly. Lawal et al. [20] and Ren et al. [21] provide reviews of methodological advances and optimization problems, as well as applications in scientific computing. The power of a PINN isn't just a neural network and a physical equation. It is dependent on the relevance of the physical residual, its correct weighting, and its numerical trainability. Inappropriate physical limitations can hinder training or bias the results.

Bearing RUL prediction is not a textbook PINN problem because the degradation process is not governed by a single known partial differential equation. Rolling contact fatigue, lubrication, load distribution, resonance, material variation, and measurement noise interact in ways that are difficult to reduce to one residual. Crack-growth laws such as Paris and Erdogan [22] show how mechanics can describe damage propagation in some contexts, but a complete bearing RUL equation for vibration features is rarely available in practice. This motivates weaker physics-informed learning: monotonic damage behavior, lifetime distributions, fault-frequency consistency, and other engineering priors.

### 2.2.5 Physics-Informed Neural Networks for Bearing Predictive Maintenance

Recent bearing-related studies have begun to use physics-informed or knowledge-informed learning. Chen et al. [23] proposed a physics-informed deep neural network for bearing prognosis with multisensory signals. Parziale et al. [24] applied PINNs to condition monitoring of rotating shafts. Herwig et al. [25] developed a signal-processing and physics-informed network for explainable bearing condition monitoring. Zhong et al. [26] used a multi-source physics-informed SincNet structure for rolling bearing fault diagnosis. von Hahn and Mechefske [27] investigated a knowledge-informed loss based on Weibull lifetime behavior.

These studies show why the approach is promising. Physics can improve interpretability, guide learning when data are limited, and discourage predictions that conflict with engineering expectations. They also show why the approach must be tested carefully. A physical prior only helps when it matches the measured data closely enough. If the prior is too weak, the model may behave like a data-only network. If it is too strong or mismatched, it may prevent the model from fitting valid degradation patterns.

## 2.3 RESEARCH GAP

The literature shows three gaps that motivate this thesis. First, many bearing RUL studies report strong results, but their preprocessing, RUL labeling, train-test split, and evaluation protocol differ. This makes comparisons difficult. Second, some studies use random sample splits that can leak run-specific degradation information into both training and testing. Third, physics-informed learning for bearing prognosis is still developing, and there is no guarantee that a weak physical prior improves prediction on complete held-out bearing runs.

This thesis addresses a narrower and reproducible question: how does a weak physics-informed RUL model behave when compared with common neural baselines on complete held-out IMS bearing runs? The study uses a local project structure, explicit feature extraction, a defined RUL target, saved tables and figures, and three train-validation-test splits. The goal is not to claim a universal best model. The goal is to test a specific physics-informed formulation and report what the completed experiment shows.

# Chapter 3

# Dataset and Research Methodology

## 3.1 INTRODUCTION

This chapter describes the dataset, signal-processing steps, RUL target construction, model definitions, experiment design, and evaluation metrics used in the study. The workflow begins with raw IMS vibration files and ends with model rankings, prediction curves, and error analysis. Figure 3.1 summarizes the local pipeline.

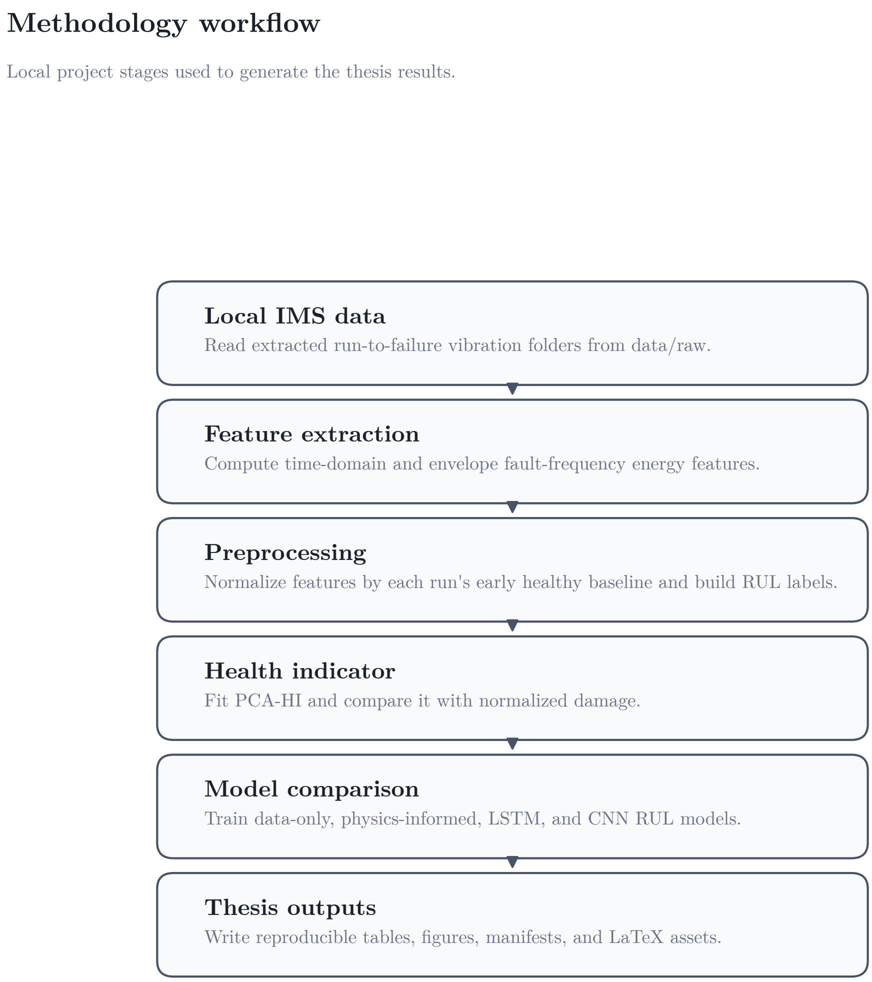


Figure 3.1: End-to-end local workflow used for IMS bearing RUL prediction.

## 3.2 EXPERIMENTAL SETUP

The IMS bearing data were generated by the NSF I/UCR Center for Intelligent Maintenance Systems with support from Rexnord Corporation. The test rig used four Rexnord ZA-2115 double-row bearings mounted on a common shaft. An AC motor kept the shaft speed constant at 2000 RPM, and a spring loading mechanism applied a radial load of 6000 lb to the shaft and bearings. The bearings were force lubricated throughout the run-to-failure tests. The IMS README also notes that the observed failures occurred after the bearings exceeded their designed lifetime of more than 100 million revolutions.

Vibration was measured at the bearing housings using PCB 353B33 High Sensitivity Quartz ICP accelerometers. The first data set used two accelerometers per bearing, placed in the x- and y-axis directions. The second and third data sets used one accelerometer per bearing. Data collection was performed with an NI DAQ Card 6062E. Figure 3.2 shows the bearing test rig and sensor placement used for the IMS experiments.

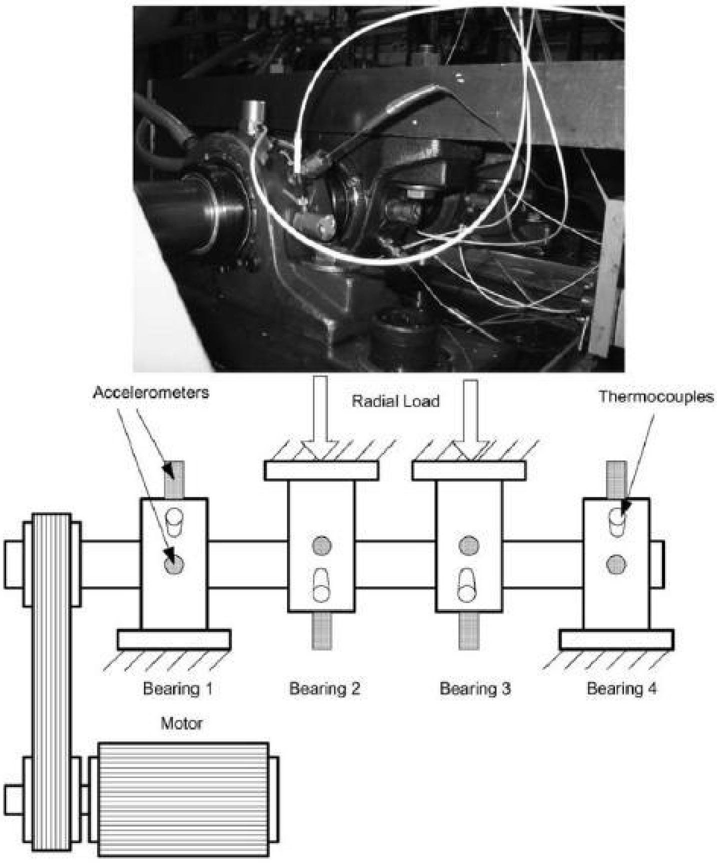


Figure 3.2: IMS bearing test rig and sensor placement used for run-to-failure data collection, adapted from the IMS Bearing Data README and Qiu et al. [1].

## 3.3 DATASET DESCRIPTION

The IMS bearing dataset contains run-to-failure vibration measurements collected from the setup described above. The raw data are stored as timestamped ASCII text files. Each file contains a nominal one-second vibration snapshot with 20,480 data points sampled at 20 kHz. The filename records the collection time, and larger gaps between timestamps indicate that the experiment resumed on the next working day. In this project, the extracted dataset folders are stored locally under `data/raw`.

The local pipeline uses three IMS folders: `1st_test`, `2nd_test`, and `3rd_test`. The first test contains eight columns because two accelerometer channels are available for each of four bearings. The second and third tests contain four columns, one for each bearing. The implementation maps each selected column to a bearing and averages two axes when the first test provides x and y directions for the target bearing.

Table 3.1: Bearing runs used in the final experiments.

| Run ID | IMS dataset | Bearing number | Number of samples | Used in experiments |
|---|---|---|---|---|
| ds2_b1 | 2nd_test | 1 | 984 | Exp 1 Train; Exp 2 Train; Exp 3 Train |
| ds1_b3 | 1st_test | 3 | 2156 | Exp 1 Train; Exp 2 Validation; Exp 3 Test |
| ds1_b4 | 1st_test | 4 | 2156 | Exp 1 Validation; Exp 2 Train; Exp 3 Validation |
| ds3_b3 | 3rd_test | 3 | 6324 | Exp 1 Test; Exp 2 Test; Exp 3 Train |

Figure 3.3 shows the number of snapshots available for each selected run. The difference in run length is large. The third dataset bearing run has 6324 samples, while the second dataset bearing run has 984 samples. This imbalance affects model training and is one reason why Experiment 3 uses balanced sampling for its training runs.

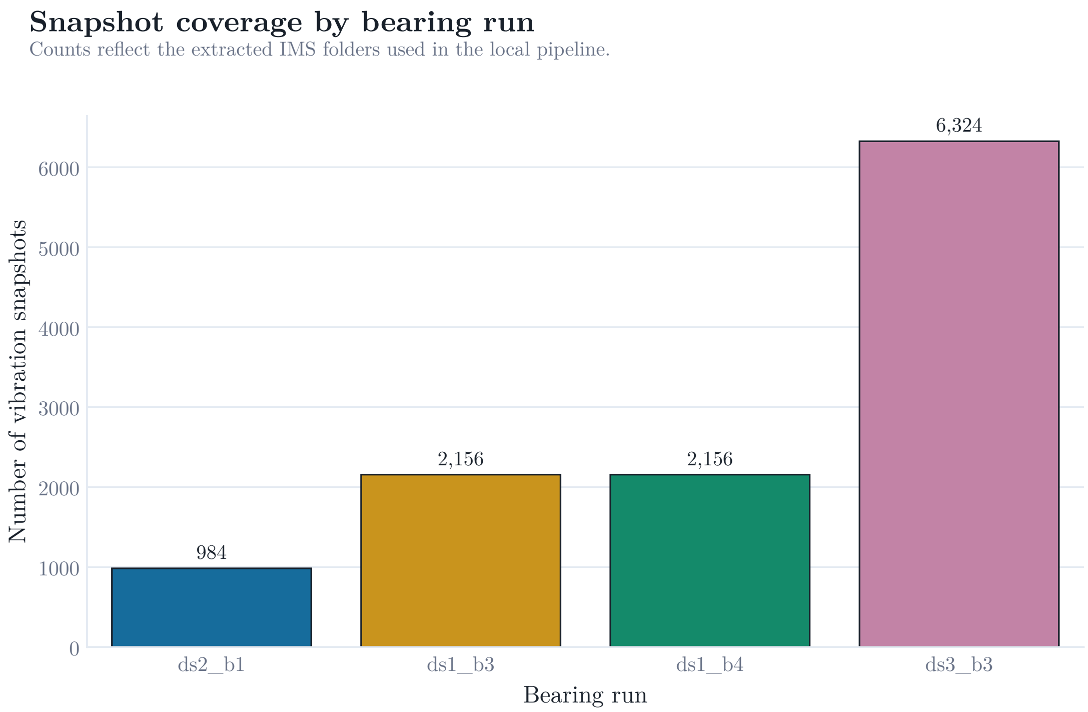


Figure 3.3: Snapshot coverage for the four bearing runs used in the final experiments.

## 3.4 FEATURE EXTRACTION

Each vibration snapshot is converted into a feature vector. The time-domain features are RMS, standard deviation, peak-to-peak value, kurtosis, crest factor, and mean absolute value. These features summarize amplitude, spread, impulsiveness, and peak behavior. They are widely used in bearing condition monitoring because they are simple and physically interpretable.

The physics-related features are calculated from envelope spectral energy near bearing fault frequencies. The signal is centered, transformed with the Hilbert transform to obtain the envelope, centered again, windowed with a Hann window, and transformed with a real FFT. For each fault frequency, the pipeline sums spectral energy in a 5 Hz band around the first four harmonics. The four energy features are $E_{FTF}$, $E_{BPFO}$, $E_{BPFI}$, and $E_{BSF}$. Their sum is stored as $E_{kin}$.

The fault-frequency equations used by the implementation are

$$FTF = \frac{f_r}{2}\left(1 - \frac{d}{D}\cos\theta\right), \tag{3.1}$$

$$BPFO = \frac{nf_r}{2}\left(1 - \frac{d}{D}\cos\theta\right), \tag{3.2}$$

$$BPFI = \frac{nf_r}{2}\left(1 + \frac{d}{D}\cos\theta\right), \tag{3.3}$$

$$BSF = \frac{Df_r}{2d}\left(1 - \left(\frac{d}{D}\cos\theta\right)^2\right), \tag{3.4}$$

where $f_r$ is shaft frequency, $n$ is the number of rolling elements, $d$ is rolling-element diameter, $D$ is pitch diameter, and $\theta$ is contact angle.

Table 3.2: Extracted features used by the neural models.

| Feature name | Domain | Physical meaning | Used by models |
|---|---|---|---|
| elapsed_scaled | time feature | Normalized elapsed operating time used as temporal input | yes |
| rms | time-domain | Signal energy amplitude through root mean square vibration | yes |
| std | time-domain | Vibration dispersion around the mean | yes |
| ptp | time-domain | Peak-to-peak vibration range | yes |
| kurtosis | time-domain | Impulsiveness of vibration signal | yes |
| crest_factor | time-domain | Peak amplitude relative to RMS | yes |
| mean_abs | time-domain | Mean absolute vibration amplitude | yes |
| E_FTF | physics-informed frequency feature | Envelope spectral energy around cage fault train frequency harmonics | yes |
| E_BPFO | physics-informed frequency feature | Envelope spectral energy around outer-race fault frequency harmonics | yes |
| E_BPFI | physics-informed frequency feature | Envelope spectral energy around inner-race fault frequency harmonics | yes |
| E_BSF | physics-informed frequency feature | Envelope spectral energy around rolling-element fault frequency harmonics | yes |
| E_kin | physics-informed frequency feature | Combined bearing kinematic fault-frequency spectral energy | yes |
| sigma_H_norm | auxiliary placeholder | Constant placeholder; not a Hertzian contact stress calculation | yes |

Table 3.3: Bearing fault frequencies used for envelope spectral-energy extraction.

| Frequency name | Symbol | Value in Hz | Related bearing component | Extracted feature |
|---|---|---|---|---|
| FTF | FTF | 14.78 | Cage / fundamental train | E_FTF |
| BPFO | BPFO | 236.40 | Outer race | E_BPFO |
| BPFI | BPFI | 296.93 | Inner race | E_BPFI |
| BSF | BSF | 139.92 | Rolling element | E_BSF |

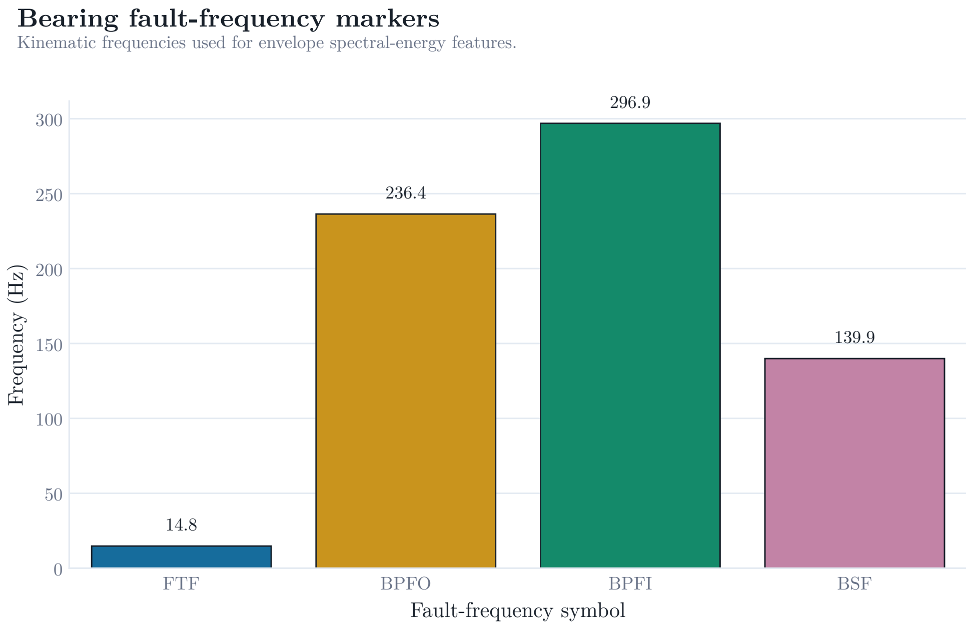


Figure 3.4: Bearing fault-frequency values used as markers for envelope spectral-energy extraction.

## 3.5 RUL TARGET LABELING

For each selected bearing run, elapsed time is computed from the timestamp of the first snapshot. RUL is computed by subtracting the current timestamp from the timestamp of the final snapshot. The model target is normalized RUL:

$$RUL_{norm}(t) = \frac{RUL(t)}{\max(RUL)}. \tag{3.5}$$

This gives a target near 1 at the beginning of the run and near 0 at the end. The normalization makes results comparable across runs with different durations. It also means that the model predicts relative life fraction, not absolute hours.

## 3.6 PREPROCESSING

Feature values can differ greatly between bearing runs. To reduce the effect of run-specific scale, each run is normalized using its early healthy baseline. The first 5 percent of samples, with a minimum of 20 samples, are treated as the healthy baseline. For each model feature, the median baseline value is used. The feature is converted to a nonnegative relative increase, transformed with `log1p`, and smoothed with a rolling median of 15 samples.

Elapsed time is converted to `elapsed_scaled` by dividing elapsed hours by 1200. The model input vector contains `elapsed_scaled`, the normalized feature set, and `sigma_H_norm`. The final variable is a constant placeholder kept for compatibility with the original notebook structure. It is not treated as a real Hertzian contact stress calculation in the final interpretation.

### 3.7 PCA HEALTH INDICATOR

A PCA-based health indicator is built to inspect whether the extracted features contain a monotonic degradation signal. The model feature matrix is standardized and projected onto the first principal component. The resulting component is scaled between 0 and 1. If the indicator direction is inverted, it is flipped so that higher values correspond to greater damage.

The health indicator is not used as the final RUL target. It is used as an explanatory tool. It helps show whether the engineered features broadly track degradation.

Table 3.4: PCA health-indicator monotonicity.

| Bearing run | Monotonicity score | Interpretation |
|---|---|---|
| ds1_b3 | 0.766 | Strongly degradation-consistent PCA-HI behavior |
| ds3_b3 | 0.864 | Strongly degradation-consistent PCA-HI behavior |

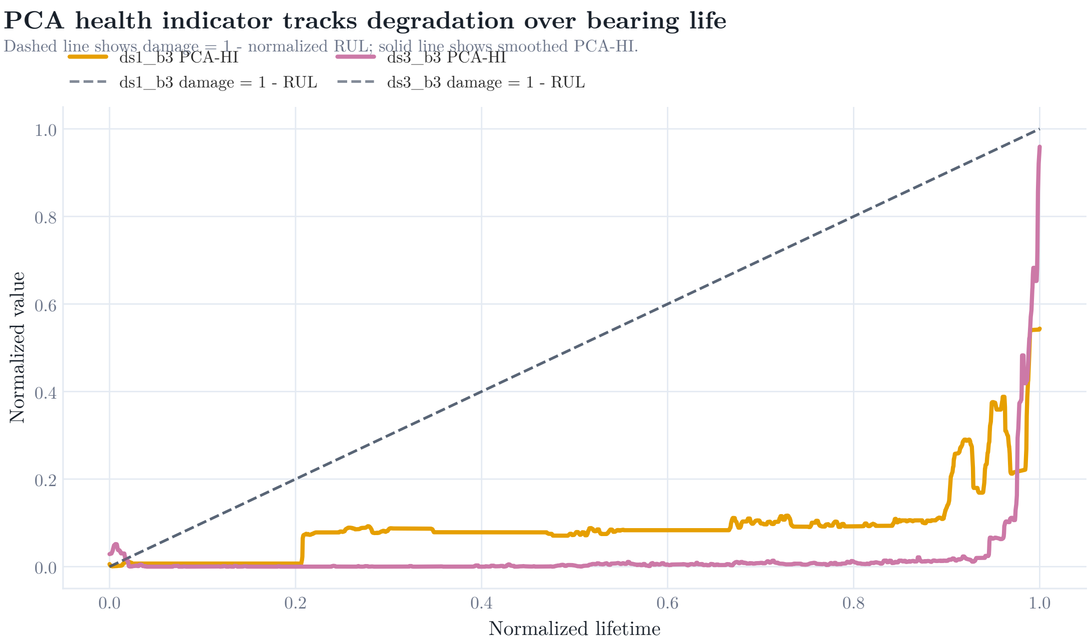


Figure 3.5: Smoothed PCA health indicator compared with normalized damage on held-out test runs.

## 3.8 MODEL DEFINITIONS

Four models are evaluated. The data-only neural baseline is a feed-forward neural network implemented in DeepXDE [28]. It uses three hidden layers with 64 neurons per layer and tanh activation. A sigmoid output transform constrains predictions between 0 and 1. It is trained with mean squared error.

The proposed physics-informed model uses the same basic feed-forward architecture but adds weak physical residuals. The first residual penalizes positive RUL slope with respect to time beyond a small tolerance, because RUL should generally decrease as life is consumed. The second residual connects high $E_{kin}$ to damage by encouraging the predicted damage term $1 - RUL$ to be consistent with fault-frequency energy. The model loss can be written as

$$\mathcal{L} = \mathcal{L}_{data} + \lambda_m \mathcal{L}_{mono} + \lambda_e \mathcal{L}_{energy}, \tag{3.6}$$

where $\mathcal{L}_{data}$ is the supervised RUL loss, $\mathcal{L}_{mono}$ is the monotonicity residual, $\mathcal{L}_{energy}$ is the fault-energy residual, and $\lambda_m$ and $\lambda_e$ are small regularization weights.

The LSTM baseline uses sliding windows of feature vectors. It reads a sequence of length 20 and predicts the RUL at the end of the sequence. The architecture has one LSTM layer with 64 hidden units followed by a small dense head and a sigmoid output. The CNN baseline uses the same sequence windows but applies 1D convolutions across the time axis.

Table 3.5: Model definitions and roles in the experiment.

| Model | Model type | Input format | Role in thesis | Target |
|---|---|---|---|---|
| Data-only neural baseline | Feed-forward neural network | Single snapshot feature vector | Data-only baseline | Normalized RUL |
| Proposed DeepXDE Physics-Informed RUL Model | DeepXDE physics-informed neural network | Single snapshot feature vector with time and fault-frequency energy features | Proposed model | Normalized RUL |
| LSTM baseline | Recurrent neural network | Sliding-window sequence of feature vectors | Sequence baseline | Normalized RUL |
| CNN baseline | 1D convolutional neural network | Sliding-window sequence of feature vectors | Sequence baseline | Normalized RUL |

## 3.9 EXPERIMENT DESIGN

The three experiments hold out complete bearing runs for validation and testing. This is stricter than randomly splitting snapshots because the model must transfer across bearing runs. Table 3.6 and Figure 3.6 show the split design.

Table 3.6: Train-validation-test split for each experiment.

| Experiment | Train runs | Validation run | Test run | Purpose |
|---|---|---|---|---|
| Exp 1 | ds2_b1 + ds1_b3 | ds1_b4 | ds3_b3 | Cross-dataset/cross-bearing transfer with ds3_b3 held out for testing |
| Exp 2 | ds2_b1 + ds1_b4 | ds1_b3 | ds3_b3 | Alternative training-bearing combination with the same ds3_b3 test run |
| Exp 3 | ds2_b1 + ds3_b3 | ds1_b4 | ds1_b3 | Cross-dataset transfer to ds1_b3 with balanced training runs |

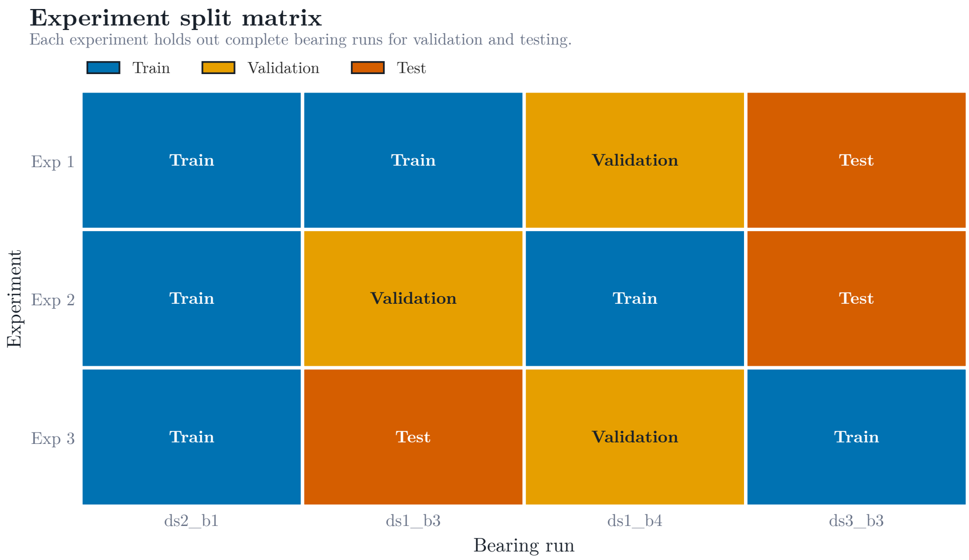


Figure 3.6: Experiment split matrix showing how each bearing run is used.

## 3.10 EVALUATION METRICS

Evaluation of the models is carried out using MAE, RMSE and $R^2$. MAE is the mean of absolute prediction errors. $R^2$ allows for larger errors to have a greater impact, since the error is squared before being averaged. The negative $R^2$ indicates that the model is worse at predicting the target than predicting the mean value of the target.

The thesis employs the RMSE as the primary ranking criterion since the errors in RUL are particularly undesirable when making decisions in maintenance planning, as RUL errors of large magnitudes are highly undesirable. Still, MAE and $R^2$ are reported to provide a complete picture of the model's behavior. The metric definitions are taken from the common practice in prognosis evaluation using the metrics described in [29].

# Chapter 4

# Local Implementation and Experimental Setup

## 4.1 LOCAL PROJECT STRUCTURE

This was the original work originally created in Google Colab notebook. The workflow was adapted to a local Python project for this thesis. Google Drive is no longer mounted in the code. It fetches data from `data/raw`, stores the extracted features in `data/processed_features` and outputs tables and figures to `outputs`.

The main package is under `src/thesis_work`. It includes individual modules for configuration, IMS data loading, feature extraction, preprocessing, model training, the metrics, report generation, and running the program from the command line. This format allows for the experiment to be easily tested and rerun, as opposed to a single notebook.

## 4.2 REPRODUCIBILITY COMMANDS

The project uses `uv` for dependency management. The main commands are:

```
uv sync --extra dev
uv run thesis-work validate-data
uv run thesis-work extract-features
uv run thesis-work run
uv run thesis-work regenerate-figures
uv run pytest -q
```

The full run trains all four models for all three experiments. The command can take a long time on CPU, so the project also supports shorter smoke-test arguments. The final results in this thesis come from the completed full run saved in `outputs/tables/final_results_table.csv`.

## 4.3 SOFTWARE ENVIRONMENT

The implementation uses Python 3.11 and the following main libraries: NumPy, pandas, SciPy, scikit-learn, matplotlib, seaborn, DeepXDE, PyTorch, and tqdm. The LaTeX thesis is compiled with MiKTeX using the Windows batch script included in the thesis source folder.

The local PC used for the completed run had an AMD Ryzen 7 5800H CPU with 8 cores and 16 logical processors, 16 GB RAM, Windows 11 Home Single Language, and an NVIDIA GeForce RTX 3060 Laptop GPU with 4 GB VRAM. Training and inference wall time is machine-specific and should be reported together with this hardware context.

## 4.4 DATA VALIDATION

The validation command checks that the expected folders are present and that the first snapshot in each dataset can be read with the correct number of columns. This step matters because the IMS text files may be stored either as extracted folders or zip files. The local loader supports both, but the final run used extracted folders under `data/raw`.

## 4.5 FEATURE CACHE

Feature extraction is the slowest preprocessing step because every vibration snapshot must be read and transformed. The pipeline therefore caches per-run feature tables and a combined `all_runs_features.csv`. Cached features make it possible to regenerate figures and tables without reprocessing the raw data or retraining the models.

## 4.6 VISUAL REPORTING

The final figures were generated with matplotlib and seaborn using a consistent color palette. Models use the same colors across all plots: gray for the data-only baseline, blue for the proposed PINN, green for LSTM, and orange-red for CNN. Bearing runs also use fixed colors. In the LaTeX thesis, plot files avoid embedded chart titles where possible because figure captions and labels provide the formal titles in the document.

# Chapter 5

# Results and Discussion

## 5.1 FEATURE BEHAVIOR OVER BEARING LIFE

The extracted features do not evolve identically across bearing runs. This is expected because different bearings can have different fault initiation times and degradation rates. RMS and peak-to-peak values capture energy growth. Kurtosis and crest factor are more sensitive to impulsive behavior. $E_{kin}$ summarizes energy near the calculated fault-frequency harmonics.

Figures 5.1 to 5.4 show representative feature evolution over normalized lifetime. The plotted feature values are normalized for visualization so that their trends can be compared across bearings.

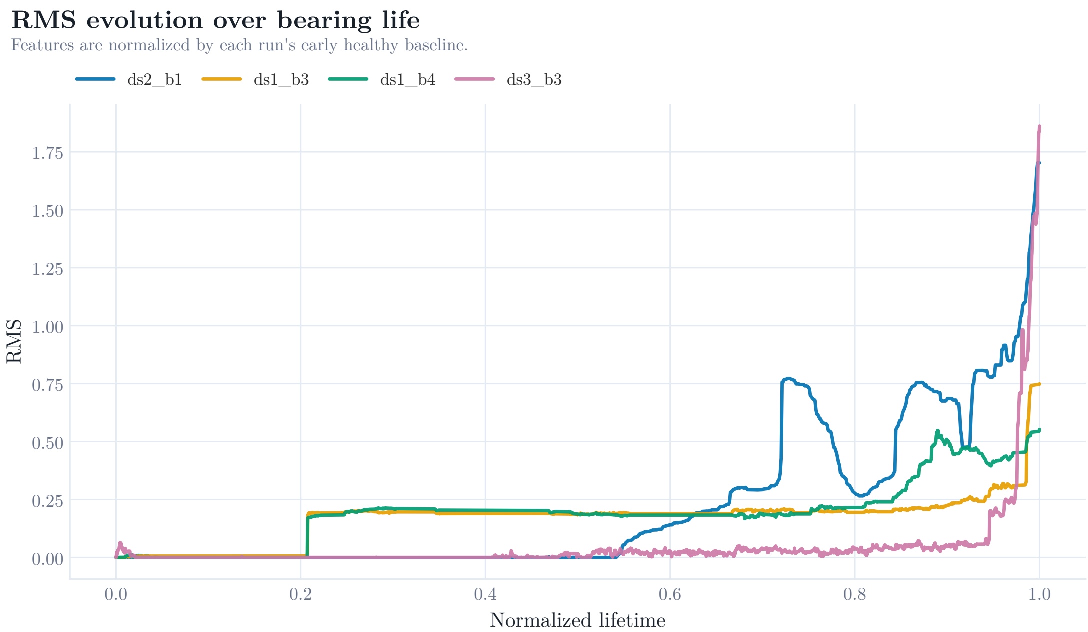


Figure 5.1: RMS evolution over normalized bearing life.

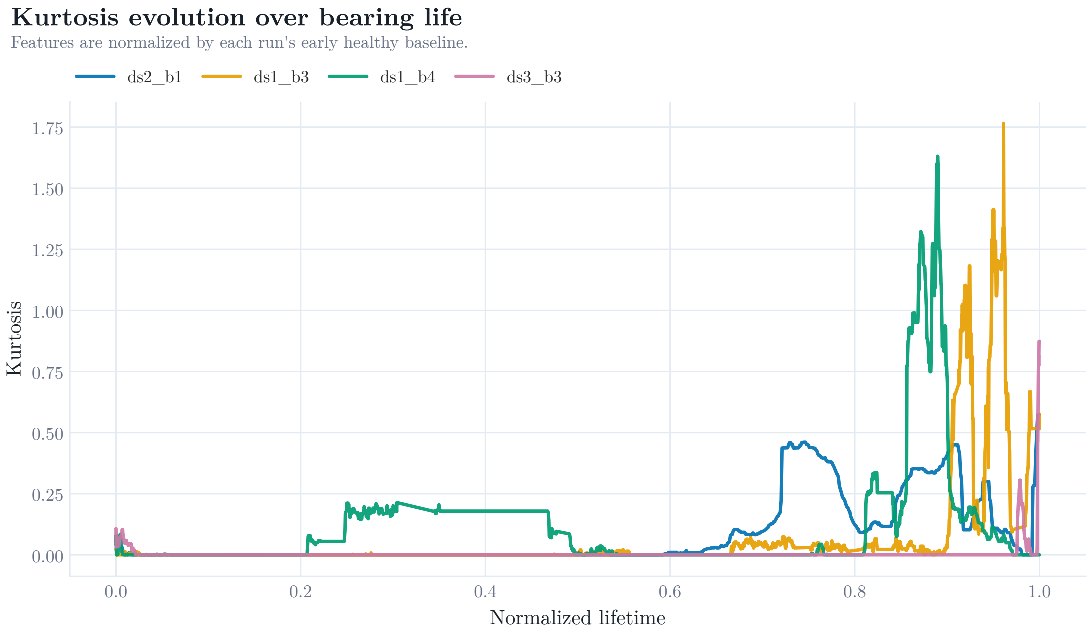


Figure 5.2: Kurtosis evolution over normalized bearing life.

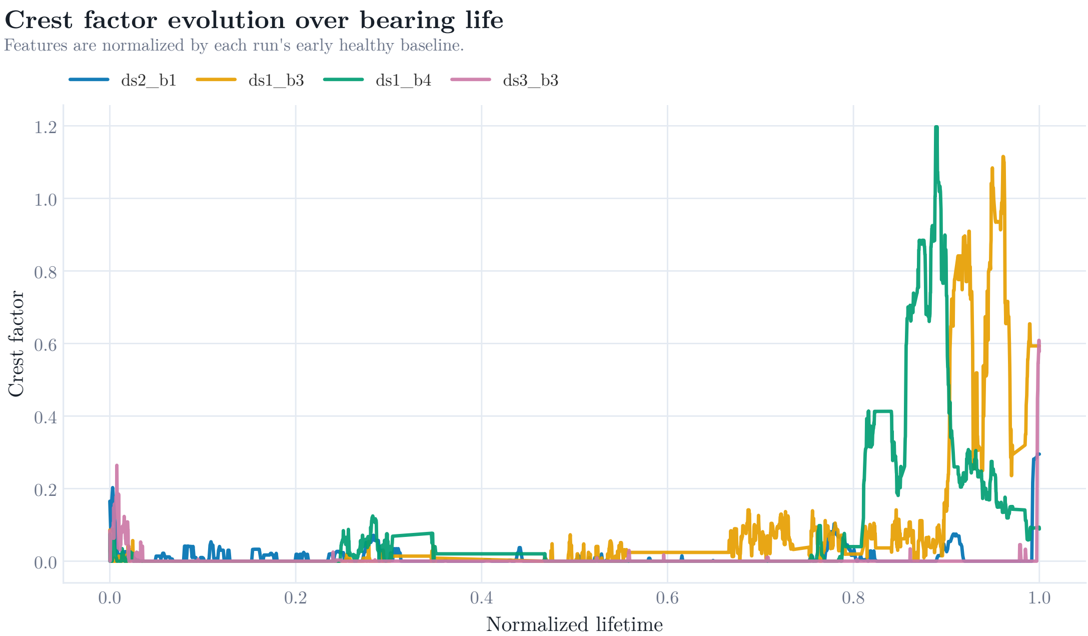


Figure 5.3: Crest factor evolution over normalized bearing life.

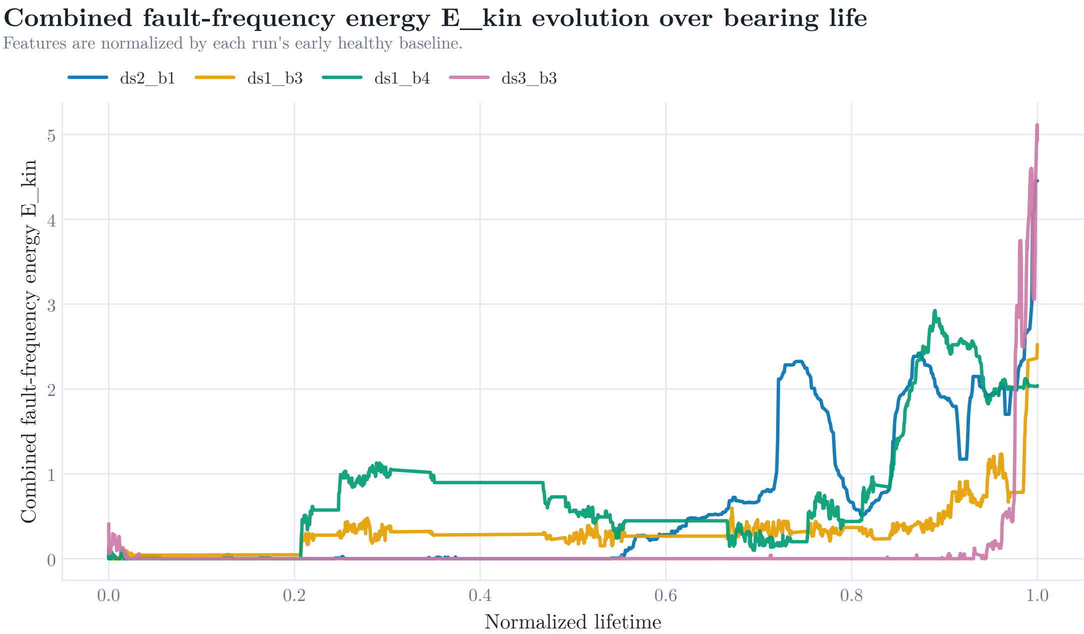


Figure 5.4: Combined fault-frequency envelope energy evolution over normalized bearing life.

The plots show why cross-bearing RUL prediction is difficult. Some features rise smoothly in one run and irregularly in another. A model trained on one degradation shape may misread a different shape. This is the central challenge behind the results that follow.

## 5.2 MAIN PREDICTION RESULTS

Table 5.1 gives the full model results for all three experiments.

Table 5.1: Final prediction metrics for all models and experiments.

| Experiment | Model | MAE | RMSE | R2 |
|---|---|---|---|---|
| Exp 1 | Data-only FNN | 0.083 | 0.128 | 0.801 |
| Exp 1 | Proposed PINN | 0.147 | 0.173 | 0.638 |
| Exp 1 | LSTM | 0.101 | 0.112 | 0.848 |
| Exp 1 | CNN | 0.244 | 0.266 | 0.146 |
| Exp 2 | Data-only FNN | 0.104 | 0.148 | 0.738 |
| Exp 2 | Proposed PINN | 0.114 | 0.144 | 0.751 |
| Exp 2 | LSTM | 0.141 | 0.155 | 0.708 |
| Exp 2 | CNN | 0.200 | 0.235 | 0.334 |
| Exp 3 | Data-only FNN | 0.156 | 0.280 | 0.032 |
| Exp 3 | Proposed PINN | 0.262 | 0.351 | -0.530 |
| Exp 3 | LSTM | 0.176 | 0.219 | 0.382 |
| Exp 3 | CNN | 0.183 | 0.204 | 0.463 |

The average performance across the three experiments is shown in Table 5.2. On average, the LSTM baseline is the strongest model. It has the lowest average RMSE at 0.162 and the highest average $R^2$ at 0.646. The data-only feed-forward baseline is second by average RMSE at 0.185. The proposed physics-informed model has an average RMSE of 0.223. The CNN baseline has the weakest average RMSE at 0.235, although it performs best in Experiment 3.

Table 5.2: Average model performance across the three experiments.

| Model | Average MAE | Average RMSE | Average R2 |
|---|---|---|---|
| CNN | 0.209 | 0.235 | 0.314 |
| Data-only FNN | 0.114 | 0.185 | 0.524 |
| LSTM | 0.139 | 0.162 | 0.646 |
| Proposed PINN | 0.175 | 0.223 | 0.286 |

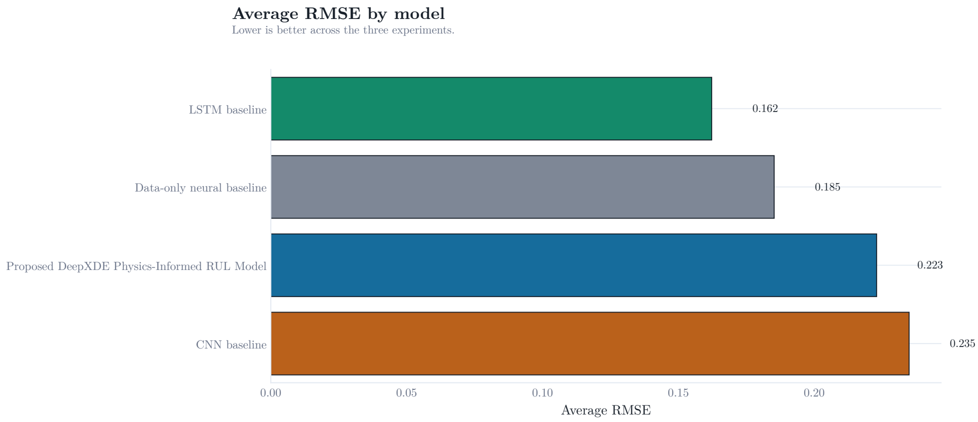


Figure 5.5: Average RMSE by model across the three experiments.

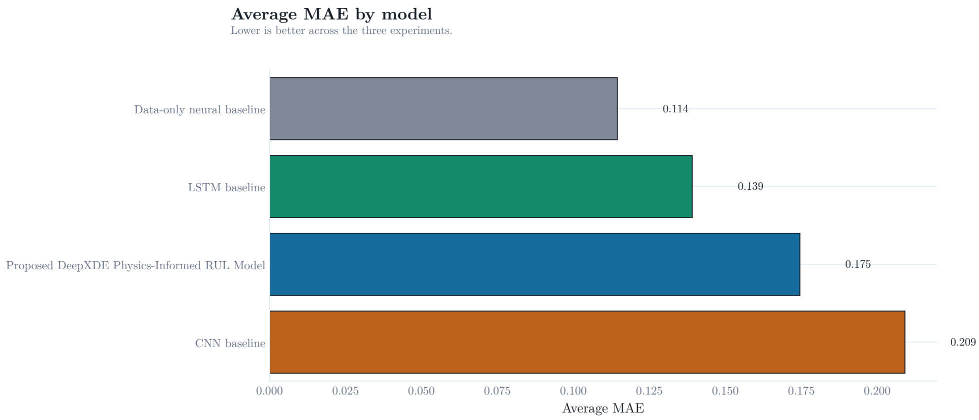


Figure 5.6: Average MAE by model across the three experiments.

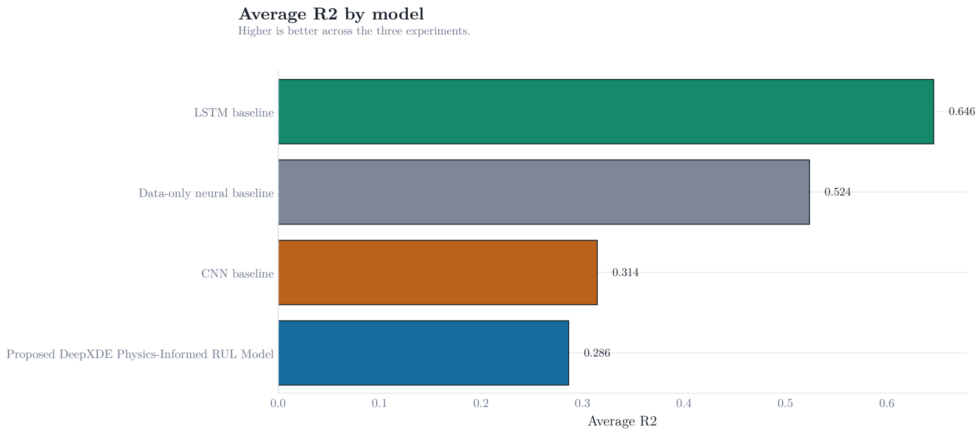


Figure 5.7: Average $R^2$ by model across the three experiments.

These results do not support the original expectation that the proposed PINN would dominate the baselines. They show a more specific conclusion: the weak physics-informed formulation can help in one split, but it is not robustly better across all held-out bearing runs.

## 5.3 EXPERIMENT-WISE COMPARISON

The model ranking by RMSE is given in Table 5.3 and Figure 5.8. The ranking figure is oriented so that ranks increase upward: rank 1 is at the bottom and rank 4 is at the top.

Table 5.3: Model ranking by RMSE.

| Experiment | Model | RMSE | Rank by RMSE |
|---|---|---|---|
| Exp 1 | LSTM | 0.112 | 1 |
| Exp 1 | Data-only FNN | 0.128 | 2 |
| Exp 1 | Proposed PINN | 0.173 | 3 |
| Exp 1 | CNN | 0.266 | 4 |
| Exp 2 | Proposed PINN | 0.144 | 1 |
| Exp 2 | Data-only FNN | 0.148 | 2 |
| Exp 2 | LSTM | 0.155 | 3 |
| Exp 2 | CNN | 0.235 | 4 |
| Exp 3 | CNN | 0.204 | 1 |
| Exp 3 | LSTM | 0.219 | 2 |
| Exp 3 | Data-only FNN | 0.280 | 3 |
| Exp 3 | Proposed PINN | 0.351 | 4 |

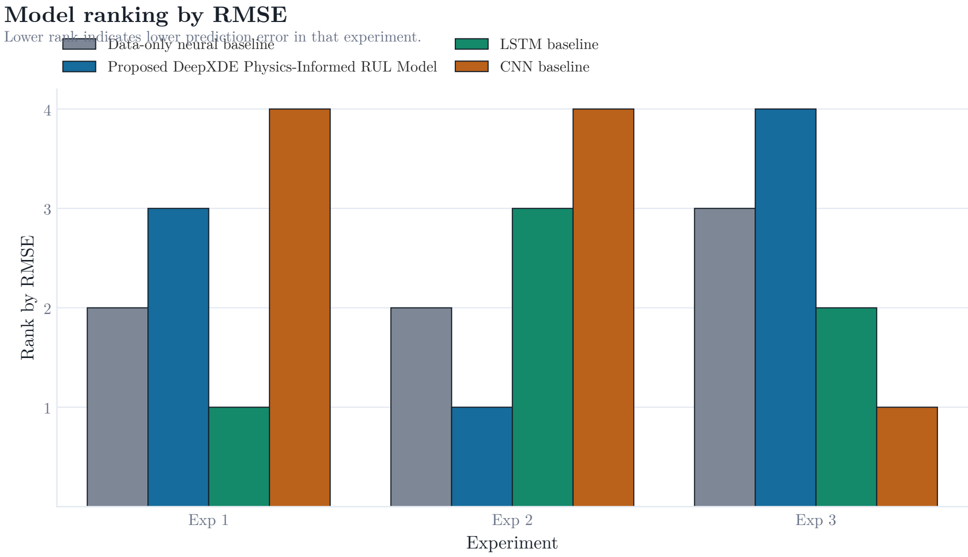


Figure 5.8: Model ranking by RMSE for each experiment. Lower rank is better.

The ranking changes from experiment to experiment. In Experiment 1, the LSTM baseline performs best with RMSE 0.112. In Experiment 2, the proposed PINN performs best with RMSE 0.144, narrowly ahead of the data-only baseline. In Experiment 3, the CNN baseline performs best with RMSE 0.204, followed by the LSTM. The proposed PINN performs worst in Experiment 3, with negative $R^2$.

Figure 5.9 makes this instability clear by showing only the lowest-RMSE model in each experiment. A single model does not win every split. The metric comparison plots in Figures 5.10 to 5.12 show the same pattern from different viewpoints.

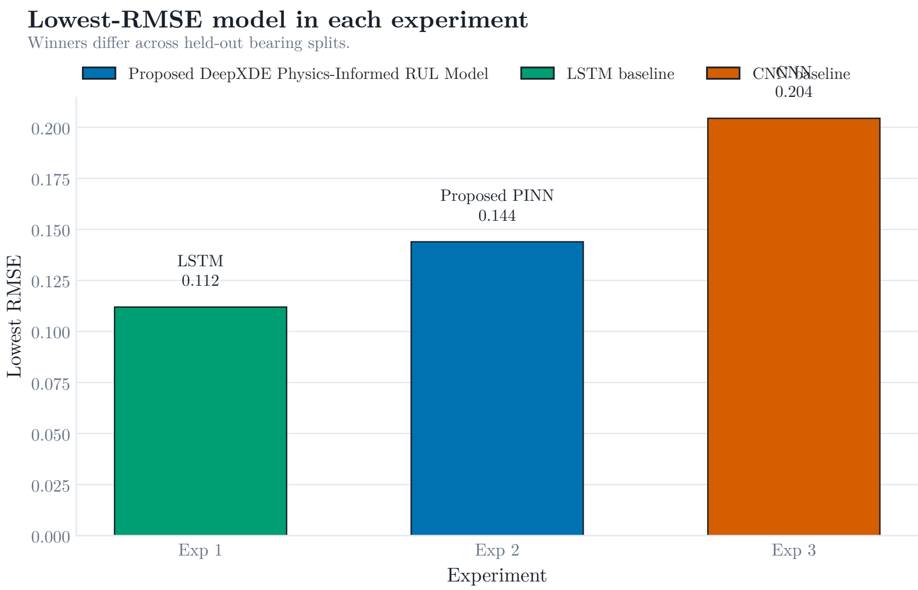


Figure 5.9: Lowest-RMSE model in each experiment.

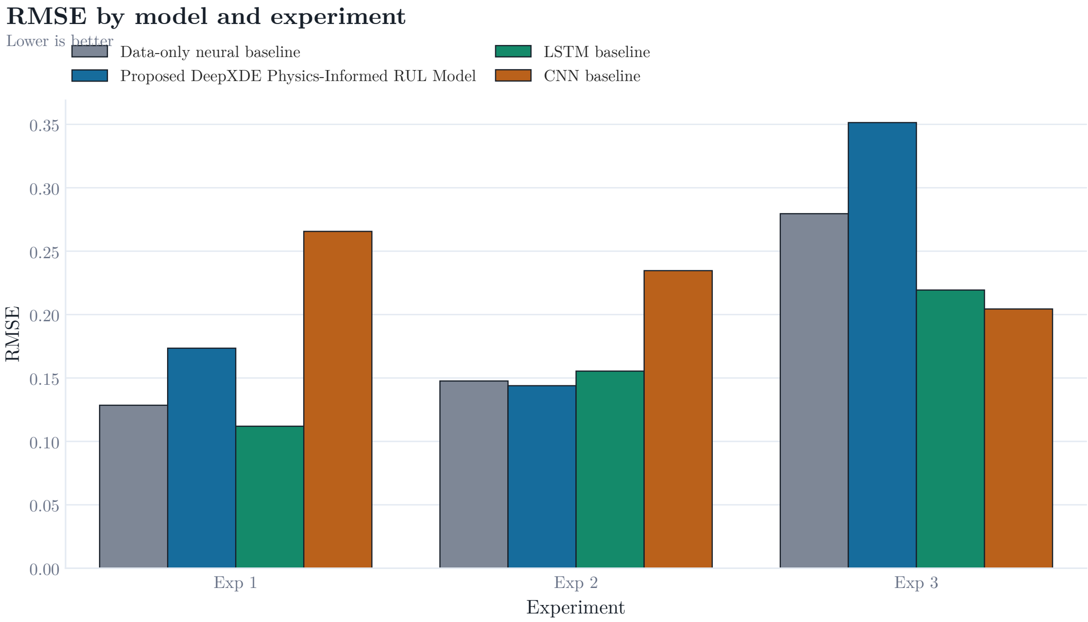


Figure 5.10: RMSE by model and experiment.

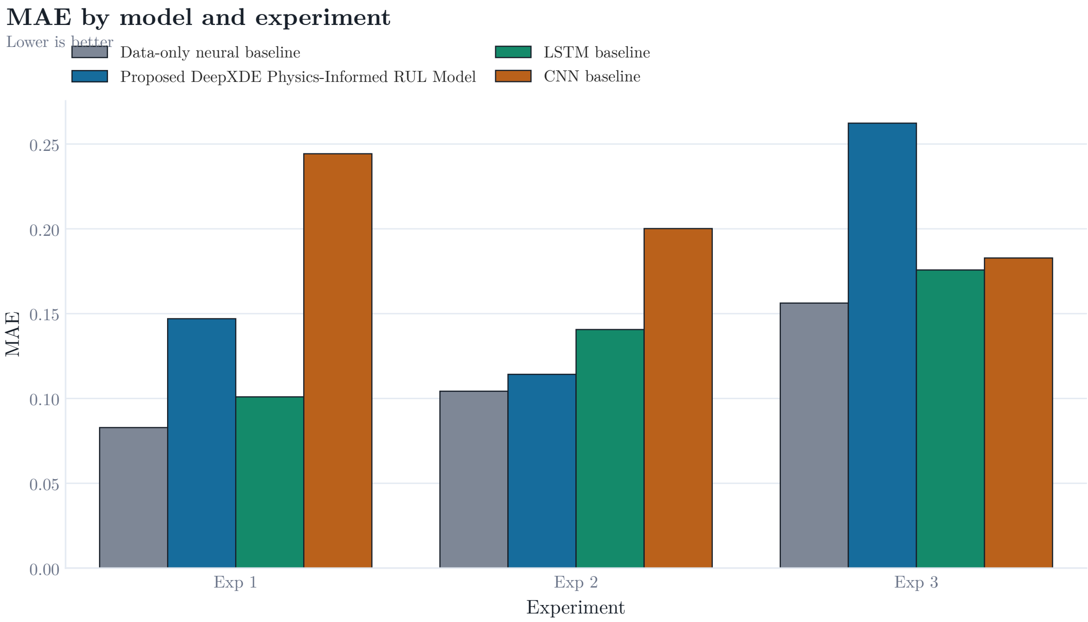


Figure 5.11: MAE by model and experiment.

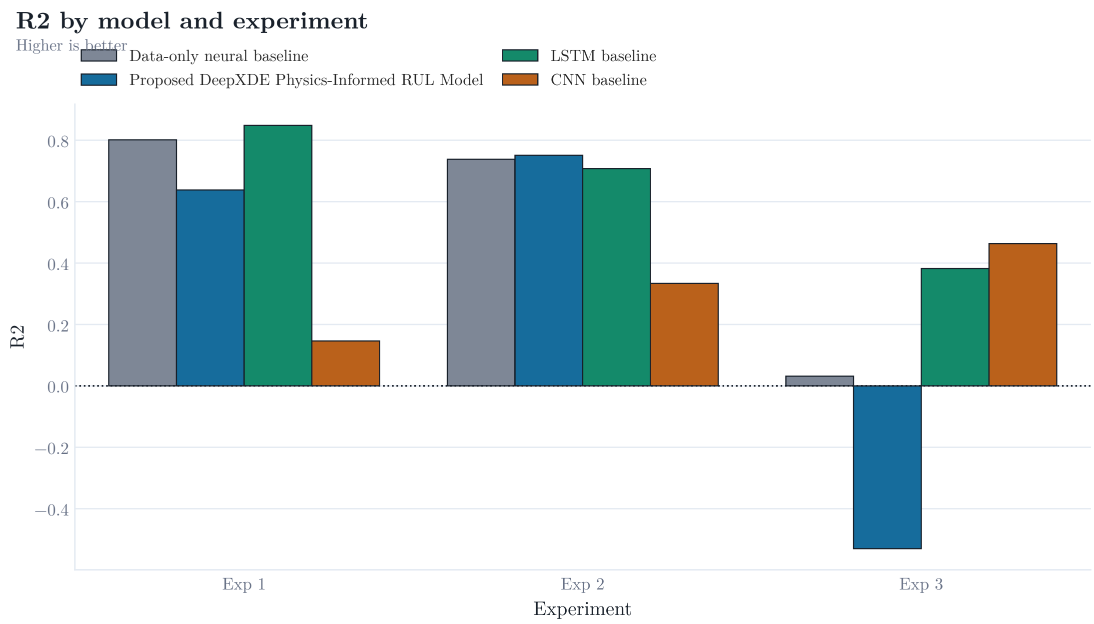


Figure 5.12: $R^2$ by model and experiment.

## 5.4 TRUE VERSUS PREDICTED RUL

Figures 5.13 to 5.15 compare the true normalized RUL curve with model predictions on the held-out test runs.

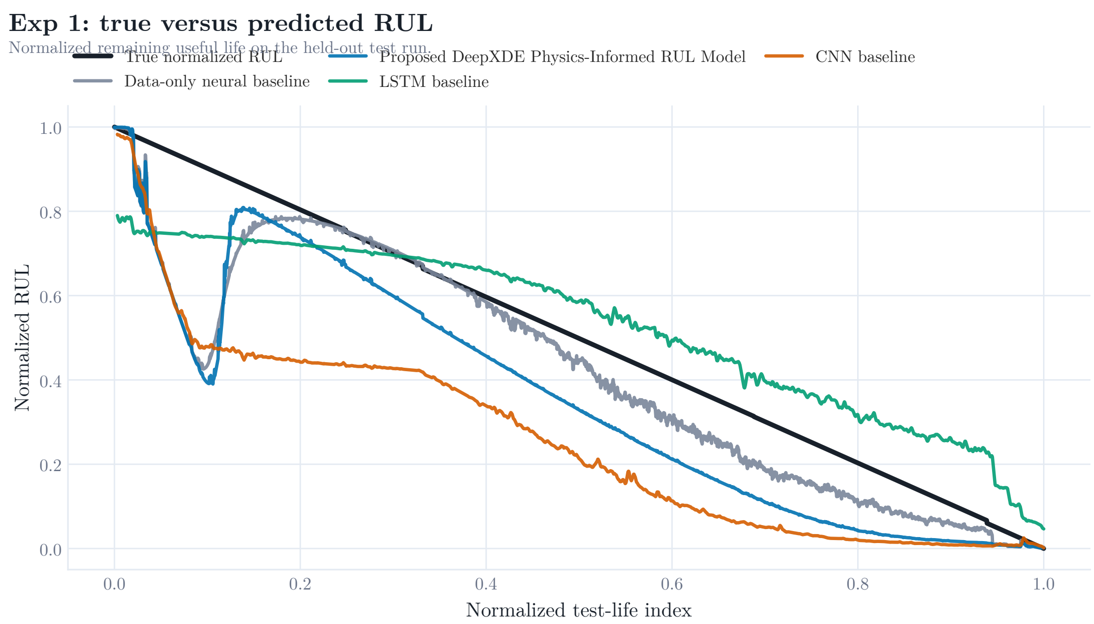


Figure 5.13: Experiment 1 true and predicted normalized RUL on the held-out test run.

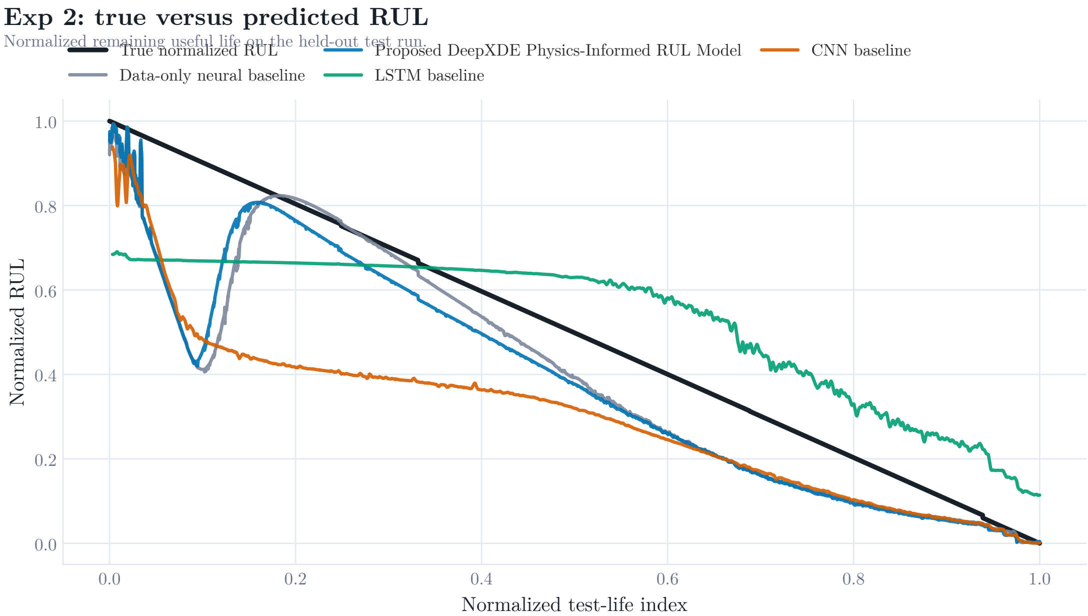


Figure 5.14: Experiment 2 true and predicted normalized RUL on the held-out test run.

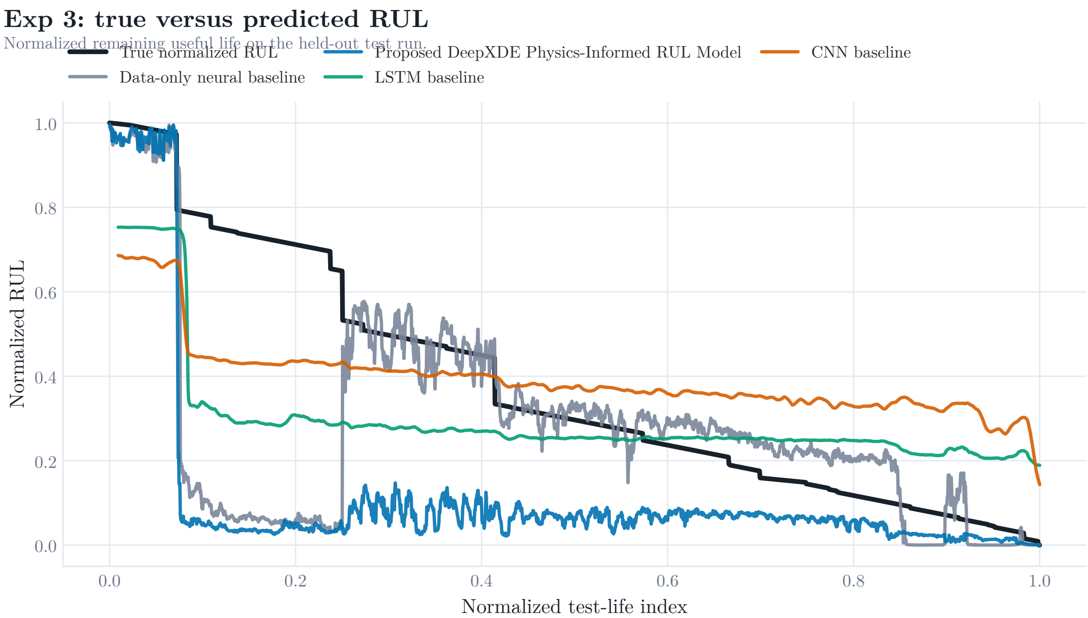


Figure 5.15: Experiment 3 true and predicted normalized RUL on the held-out test run.

In Experiment 1, the LSTM follows the decreasing RUL trend more closely than the other models. The proposed PINN is smoother, but it does not match the test curve as well. In Experiment 2, the PINN is more competitive and gives the lowest RMSE. This suggests that the physics prior can help when the training bearings give a degradation pattern compatible with the held-out test bearing. In Experiment 3, the PINN struggles on `ds1_b3`, while the CNN and LSTM sequence models handle the test run better.

## 5.5 ERROR ANALYSIS

Figure 5.16 shows smoothed absolute prediction error over normalized test life. The largest errors tend to appear when the degradation curve changes more sharply or when a model predicts RUL too conservatively near the end of life.

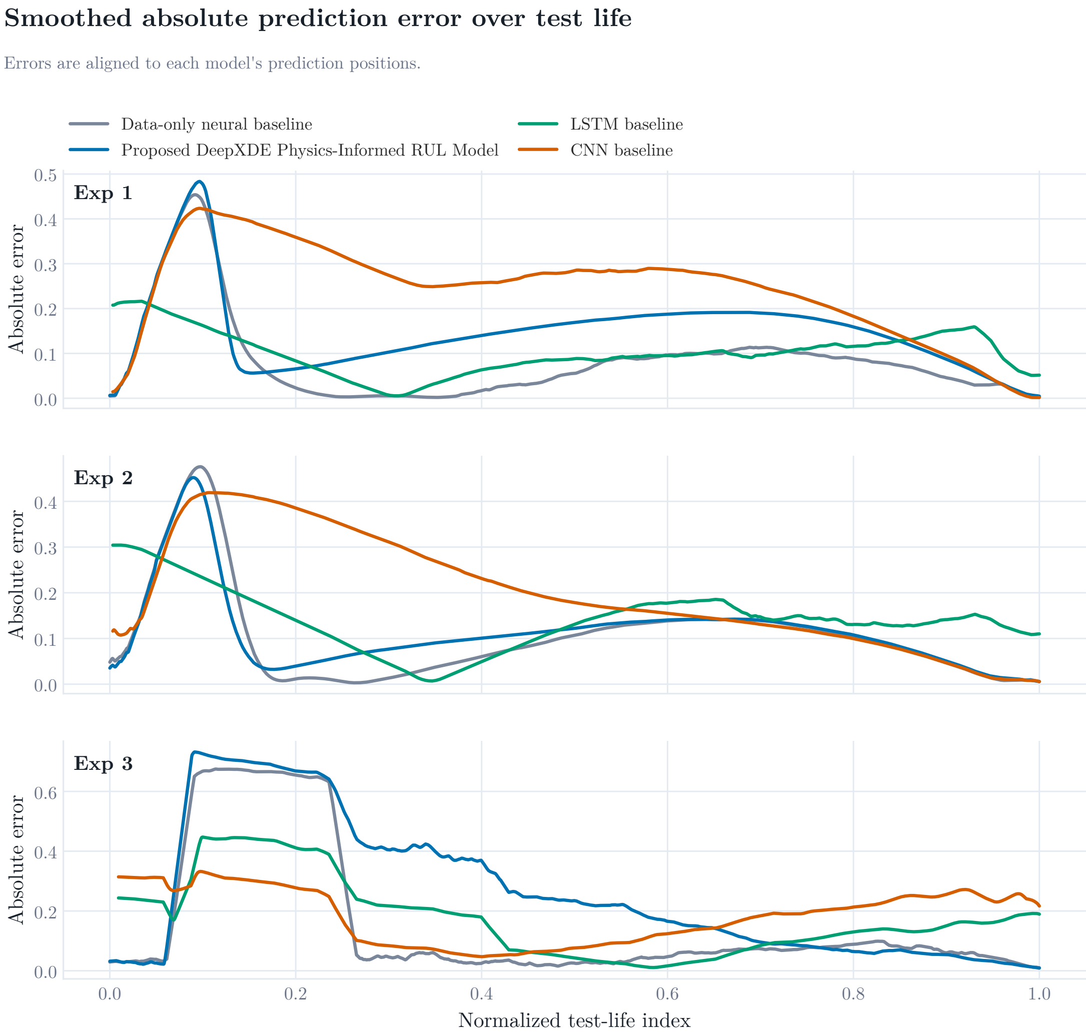


Figure 5.16: Smoothed absolute prediction error over normalized test life.

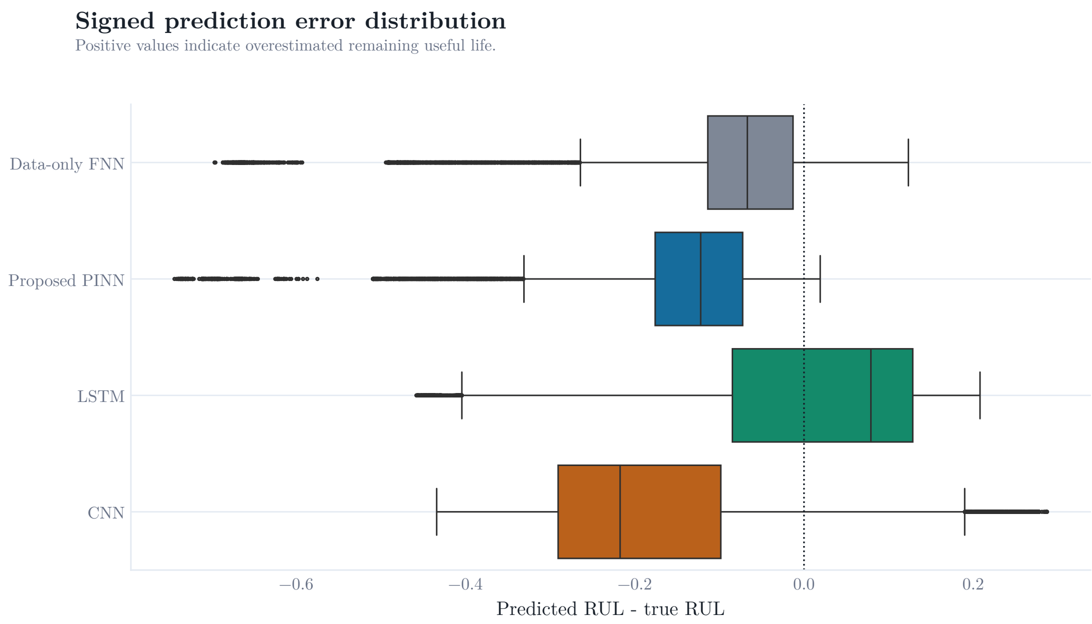


Figure 5.17: Signed prediction error distribution. Positive error means the model overestimated normalized RUL.

The distribution confirms that the sequence models do not merely improve the average score. They also reduce the spread of errors in several cases. The proposed PINN has a wider error distribution because its weak physical prior does not fully capture the different degradation paths of the test bearings.

## 5.6 DISCUSSION OF THE PHYSICS-INFORMED MODEL

The proposed PINN includes two forms of physical regularization: a monotonicity residual and a spectral prior based on fault-frequency energy. Both are reasonable engineering assumptions, but both are incomplete.

The monotonicity residual assumes RUL should not increase with elapsed life. That is correct at the life-label level, but the model sees noisy features. If the feature pattern of a held-out bearing differs from the training bearings, a strict preference for smooth monotonic behavior can prevent the model from adapting to local signal changes. The implementation uses a tolerance and small weight to reduce this issue, but the result still depends on the split.

The spectral prior assumes that higher combined fault-frequency energy should correspond to greater damage. This is a plausible thing physically, but envelope energy is not an actual damage type. It may be detected by resonance, noise, axis direction and load-zone behavior as well as the occurrence of fault development. Because $E_(kin)$ growth is late or late sporadically in one run, a model learned from another run could use the previous on the wrong time.

The take-away message from the strongest lesson is that physics-informed learning is not necessarily superior just because it uses engineering terminology. The previous one must be close enough to the data generation process to assist. In this thesis the former is helpful in the case of Experiment 2 but not sufficient to overcome sequence modeling on the whole.

## 5.7 WHY THE LSTM PERFORMED WELL

While the LSTM baseline is based on recent history, not a single snapshot. This is significant as degradation is a process. The trend of a feature value provides additional information; a single value may be ambiguous. Some examples of moderate RMS may be early healthy operation in one run, or a temporary plateau upon fault growth in another. A brief sentence is provided to make this distinction.

The LSTM also contains much less assumptions regarding the exact physical link between $E_(kin)$ and damage. It directly learns temporal patterns from the feature sequences. This is likely to be the reason for its best average RMSE and $R^2$.

## 5.8 WHY THE CNN WON EXPERIMENT 3

The CNN baseline is the least successful overall, while it is the most successful in Experiment 3. This outcome indicates that the choice of the model is affected by the pairing of the train and test sets. A 1D CNN is able to recognize the temporal patterns within the sliding feature window. Those local patterns also worked well for `ds1_b3` in Experiment 3. In the other experiments, the same model was not as generalized.

This is not to say the CNN is the best overall. It means that there may be behaviour in a single average metric that is not present in the split. The split-wise analysis is, therefore, as significant as the average table in the case of a thesis.

## 5.9 PRACTICAL IMPLICATIONS

In case of an engineering maintenance system, the best and safest conclusion is the conservative one. The current PINN formulation should not be deployed as a final RUL predictor. The LSTM baseline is stronger in the completed experiments. But it also needs more validation before field use. A practical system is going to need more run-to-failure data, multiple operating conditions, uncertainty estimates along with a clear maintenance decision rule.

The work still provides a useful direction. Physics-informed features are important for interpretation, and the PINN framework is promising. The future work to be done is not to abandon physics and improve the physical model, use better degradation constraints, and evaluate on more splits and datasets.

# Chapter 6
# Conclusion

## 6.1 SUMMARY

In this thesis the we investigated the prediction of remaining useful life for rolling element bearings. The study used the IMS run-to-failure dataset. To ensure the workflow is functional without Google Drive, we converted the original notebook into a local Python project. By using this pipeline, the system reads raw dataset folders and extracts features from the time domain and the envelope spectrum. As part of the process, the system creates normalized labels for remaining useful life plus develops a health indicator using principal component analysis. For the final stages, the software trains four neural models and generates tables and figures that can be reproduced.

The proposed model is a physics informed neural network within the DeepXDE framework. It uses weak physical residuals that relate to the energy at fault frequencies but also the fact that remaining useful life decreases over time. To evaluate this model, the researcher compared it with a feed forward neural baseline that uses only data, an LSTM sequence baseline and a CNN sequence baseline - those comparisons occurred during three experiments involving different bearings.

here were different outcomes across the various tests - in Experiment 2, the proposed physics informed neural network resulted in the lowest root mean square error. It is not the most effective model when considering all data. The LSTM baseline is the most consistent performer, as it has an average root mean square error of 0.162 and an average coefficient of determination of 0.646. If ranked by average root mean square error, the data only baseline is the second most effective model. The proposed physics informed neural network is in third place according to this metric. And the CNN is in fourth place, although the CNN is the most accurate model in Experiment 3.

## 6.2 CONTRIBUTIONS

The main contributions of this thesis are:

1. A local RUL prediction pipeline for the IMS bearing dataset.

2. A feature extraction method combining time-domain vibration features and bearing fault-frequency envelope energy which has been documented.
3. A degradation consistency for analysis of PCA-based health indicator.
4. A DeepXDE physics-informed RUL model using monotonicity and frequency features.
5. A comparison with feed-forward, LSTM, and CNN baselines under complete held-out bearing splits.
6. A discussion of the mismatch between the original proposal expectation and the final experimental results.

### 6.3 LIMITATIONS

The study consists of a couple limitations. The number of complete failure runs were low. The RUL target is normalized by the final timestamp, so it does not represent an independently measured failure threshold. The physical prior is weak and does not include a full contact fatigue model, lubrication model, temperature effect, or uncertainty estimate. The model is trained on engineered features rather than raw waveforms. Finally, the results are from one local implementation and one set of hyperparameters.

### 6.4 FUTURE WORK

Future work should test stronger physics-informed formulations. Possible directions include a learned degradation state constrained by monotonic damage growth, uncertainty-aware RUL prediction, Weibull or crack-growth inspired lifetime priors, and multi-task learning that predicts both health indicator and RUL. The experiments should also be repeated on PRONOSTIA and other bearing datasets so that conclusions do not depend on one benchmark.

The feature pipeline can also be improved. Time-frequency representations, adaptive envelope bands, bearing-specific resonance selection, and raw waveform sequence models may capture information lost by the current feature set. More careful hyperparameter tuning and repeated random seeds would make the comparison more statistically reliable.

### 6.5 FINAL CONCLUSION

The completed experiments show that physics-informed bearing RUL prediction is promising but not automatic. In this thesis, the proposed PINN improved the result in one experiment but did not beat the LSTM baseline overall. The best model was the one that used temporal context most effectively. A practical bearing prognostics system should therefore combine physical interpretation with sequence-aware learning, and it should be tested under complete run-level splits before strong claims are made.

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# Chapter A

# Reproducibility Notes

## A.1 FINAL RESULT TABLE

The main result table is repeated here for quick reference.

Table A.1: Final prediction metrics for all models and experiments.

| Experiment | Model | MAE | RMSE | R2 |
|---|---|---|---|---|
| Exp 1 | Data-only FNN | 0.083 | 0.128 | 0.801 |
| Exp 1 | Proposed PINN | 0.147 | 0.173 | 0.638 |
| Exp 1 | LSTM | 0.101 | 0.112 | 0.848 |
| Exp 1 | CNN | 0.244 | 0.266 | 0.146 |
| Exp 2 | Data-only FNN | 0.104 | 0.148 | 0.738 |
| Exp 2 | Proposed PINN | 0.114 | 0.144 | 0.751 |
| Exp 2 | LSTM | 0.141 | 0.155 | 0.708 |
| Exp 2 | CNN | 0.200 | 0.235 | 0.334 |
| Exp 3 | Data-only FNN | 0.156 | 0.280 | 0.032 |
| Exp 3 | Proposed PINN | 0.262 | 0.351 | -0.530 |
| Exp 3 | LSTM | 0.176 | 0.219 | 0.382 |
| Exp 3 | CNN | 0.183 | 0.204 | 0.463 |

## A.2 COMMANDS USED FOR THE LOCAL PIPELINE

```
uv sync --extra dev
uv run thesis-work validate-data
uv run thesis-work extract-features
uv run thesis-work run
uv run thesis-work regenerate-figures
uv run pytest -q
```

## A.3 RAW DATA LOCATION

The raw IMS folders are expected in:

```
data/raw/
  1st_test/
  2nd_test/
  3rd_test/
```

These folders are intentionally not part of the LaTeX source because they contain the local raw dataset.

## A.4 LATEX BUILD

The LaTeX source is stored under `thesis/latex`. The file `build_pdf.bat` compiles the document into `physics_informed_bearing_rul_thesis.pdf` using MiKTeX.